1. POWER / USB

J1 -> U8 (ESD) -> +5V -> U7 (LDO) -> +3V3 ; SW1 selects VREF = +3V3 / EXT / +5V

2. MCU

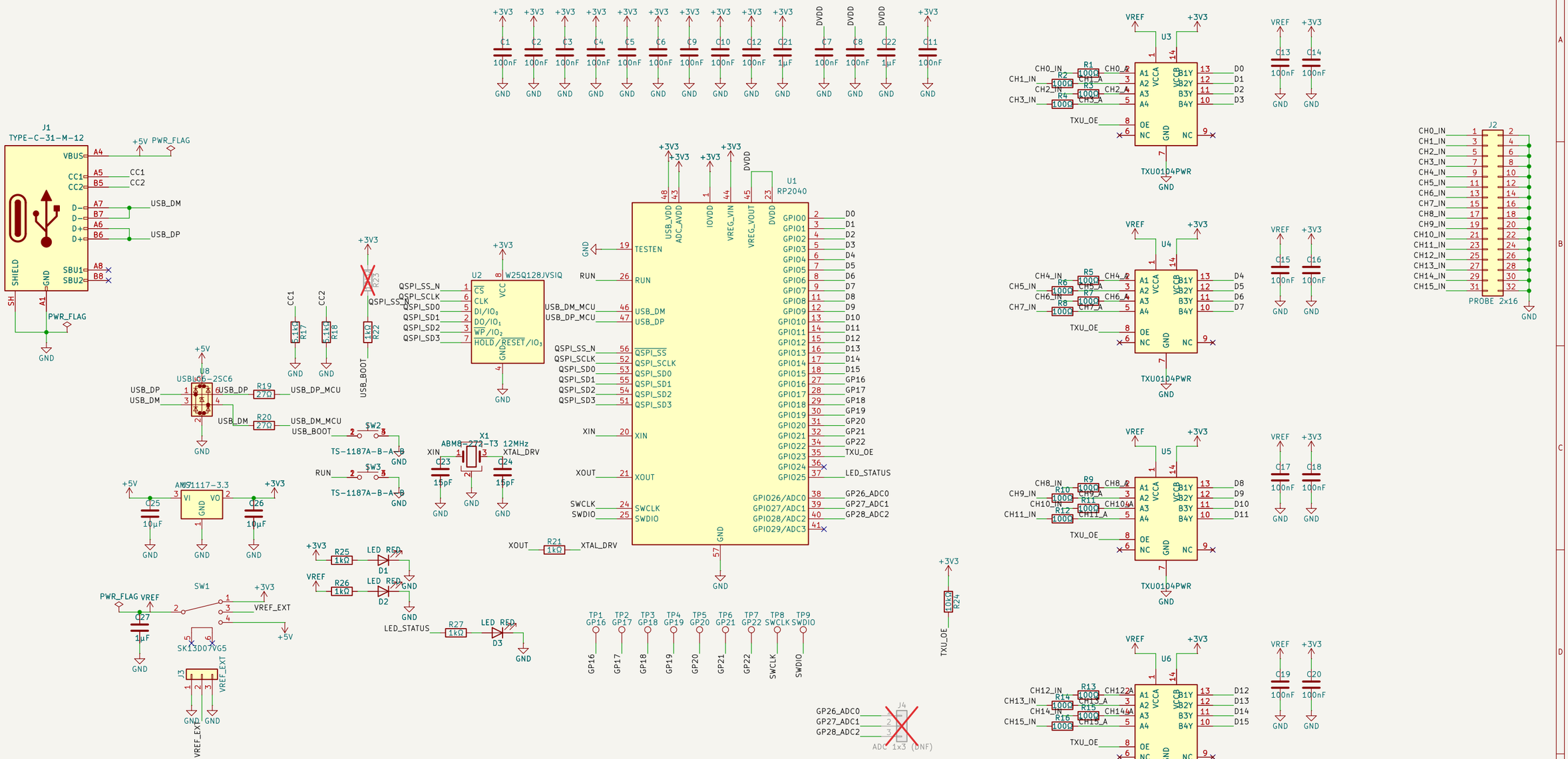
U1 RP2040 + U2 W25Q128 (QSPI) + X1 12MHz ; 100nF x10 (C1–C10) + 1uF x2 (C21/C22) ; BOOT = SW2, RUN = SW3 ; SWD on TP8/TP9

3. FRONT END x4

100R -> TXU0104 (A: VREF, B: +3V3) -> Dn ; OE = TXU_OE

4. PROBE HEADER

J2 2x16: odd = CH0–CH15, even = GND



Generated by tools/gen_schematic.py from SPEC.md / KICAD-KIT.md (2026-09-20). Placement is functional only; tidy in the GUI.

SW1 (SK13D07VG5) is a SHORTING slide switch: +3V3 (pin 1) and +5V (pin 4) are never adjacent; EXT (pin 3) sits between. Do not slide with EXT connected.

USB path: J1 -> U8 (USBLC6) -> R19/R20 27R -> RP2040. No external pull-ups (RP2040 internal). CC1/CC2: 5.1k each to GND.

Crystal: 1k damping on the XOUT side (R21), XIN direct. 15pF x2 at the crystal pins. RP2040 TESTEN tied to GND (datasheet).

DNF: R23 (10k QSPI_SS pull-up), J4 (analog header). SWD on test pads TP8 (SWCLK) / TP9 (SWDIO). C27 = 1uF bulk on VREF.

Generated by tools/gen_schematic.py from SPEC.md / KICAD-KIT.md. CERN-OHL-P-2.0

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File: rp2040-logic-analyzer.kicad_sch

Title: RP2040 16ch Logic Analyzer

Size: A3	Date: 2026-09-20	**Rev: A0**
KiCad E.D.A. 10.0.3		Id: 1/1